

Attention-Guided Residual Network for Skin Lesion Classification Using Deep Reinforcement Learning

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Abstract—The increasing incidence of skin cancers emphasizes the critical need for early detection, making effective therapeutic intervention possible. Although existing skin lesion segmentation methodologies offer some solutions, they are often hindered by inconsistent annotations present in the datasets. Accurate delineation of lesion boundaries is fundamental for diagnosing various skin conditions. In this research, a skin lesion segmentation method is introduced using an attention-guided reinforcement learning algorithm, framed within the context of a Markov Decision Process. This approach mirrors the method doctors use when identifying areas of concern on the skin. To improve classification accuracy, a self-attention mechanism is integrated into the network. This addition refines the focus on vital regions within the images, ensuring more precise segmentation. With deep reinforcement learning, the agent is trained to segment these regions, employing a series of actions that are continuously variable. This continuous action space is exploited by the segmentation model, which utilizes the deep deterministic policy gradient algorithm, allowing for iterative refinement from broader segmentation to detailed delineation. The proposed method is tested on benchmark dataset, ISIC 2017, and achieved an accuracy of 97.10% for naevus, 96.33% for melanoma, and 95.13% for seborrheic keratosis cases. Additionally, the enhanced approach with attention module, outperformed several contemporary skin lesion segmentation techniques in terms of performance.

Index Terms—self attention, deep reinforcement learning, policy gradient, lesion segmentation

I. INTRODUCTION

The human body's most expansive organ is the skin. When skin cells grow without restraint and order, it results in skin cancer, which can swiftly spread. One of the most lethal types of skin cancer is melanoma, and its occurrences have shot up over the decades [1]. Detecting it early boosts survival chances. Dermoscopy imaging is used to spot melanoma or potential skin issues by observing pigmented skin irregularities. This method is non-intrusive and can detect lesions early. Der-

moscopy images [2] have high clarity, allowing dermatologists to inspect skin lesions visually. CNN has shown potential in automating skin lesion assessments, similar to dermatologists, despite the time-consuming, expertise-intensive, and subjective nature of traditional skin lesion assessments. Automated skin lesion analysis can enhance computer-aided diagnosis in dermatology clinics by incorporating clinical context and using contextual images from the same patient. The ABCD [3] guidelines outline specific criteria for melanoma diagnosis, including irregular edges, skin color changes, and lesions exceeding 6mm. However, automation of segmentation remains unresolved. Some lightly pigmented skin lesions can have similar visual patterns to surrounding skin, leading to vague boundaries and complicating segmentation. Figure 1 presents skin lesion images from the International Skin Imaging Collaboration (ISIC-2017) [4] collection with noticeable class differences stemming from blurred borders, non-uniformity, and the presence of hairs. High-resolution dermoscopic images require significant processing time and resources, making identification of lesion boundaries challenging due to intricate textures and different features from various medical centers. Multi-scale data in CNN structures or multi-task learning models aim to leverage data for reliable predictions but often add complex training parameters or require more labeling details.

In methods employing fully convolutional networks (FCNs), the FCN tends to simplify intricate structures and overlooks finer details. Stand-alone post-processing utilizes CRF, which operates separately from the FCN. Rapid shifts between down-sampling sequences and increased sampling rates often lead to detail loss and subpar segmentation outcomes. Oversized objects frequently pose challenges, with the network making misclassifications. Even though the multi-scale method has

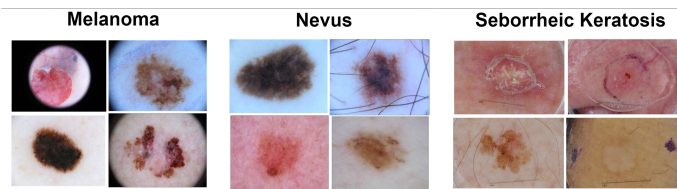


Fig. 1. Images of skin lesions from the ISIC 2017 dataset exhibit a variety of class distinctions due to several variations.

significant representational power, it leans on skip connections from low-level encoder features, which sometimes results in redundant data utilization. Additionally, when pairing the initial encoder feature's contextual information with the corresponding high-level decoder feature map, there's a deficiency in pixel-wise recognition accuracy. The process of frequent downsampling combined with high sampling rates erodes detailed information, leading to coarse lesion segmentation. This can result in mislabeling or even overlooking large objects. Integrating a self-attention network could enhance the focus on important regions and details.

Recent studies have explored the application of Deep Reinforcement Learning (DRL) in the domain of skin lesion segmentation [5]. In a lesion image, doctors typically begin with a general segmentation to capture the region of interest. This is refined through a phased approach, incorporating interactive segmentation and historical data collection. The method operates as a multi-phase segmentation procedure, automatically assimilating past knowledge. The segmentation mask serves as the baseline for the current phase, with the agent performing actions based on the current segmentation mask and input image. The methodology draws inspiration from brushwork-based techniques, integrating a self-attention mechanism to apply neural network-driven brushstrokes to the initial segmentation mask, emphasizing the region of interest and translating continuous action parameters. The continuous action parameters define the precise position and form for detailed segmentation. The Deep Deterministic Policy Gradients (DDPG) algorithm is used for continuous action space, harnessing off-policy data and Bellman's equation for policy learning. In the proposed work, the skin lesion image segmentation is considered as a Markov Decision Process (MDP) and tackled it using the DDPG algorithm. This method reflects the techniques employed by clinicians to identify and demarcate regions of interest in lesion images. Furthermore, the integration of a self-attention network into the system, has significantly improved the precision of the classification task. In addition, the model utilizes an optimized experience replay memory (ERM) to facilitate the efficient training of segmentation agent. This replay memory ensures that the model effectively recycles past experiences, fostering multiple avenues of learning. Finally, the resilience and efficiency of the skin lesion classification approach have been showcased, comparing the results against other leading-edge techniques in the dermatological domain.

II. RELATED WORKS

Medical image segmentation has been profoundly impacted by deep learning, which provides reliable solutions for difficult tasks including the detection and analysis of complicated disorders. The complexity of medical images has proven challenging for traditional image analysis techniques, particularly as diseases reveal themselves in more complex ways. In their thorough evaluation of various deep learning techniques for medical picture segmentation, As in [6] it highlights the improvements and difficulties of structures like CNNs, 2.5D and 3D approaches, U-Net [7], SegNet [8] and RNNs. However, Mittal et al. [9] focus on the diagnosis of brain tumours and investigate the possibility of deep learning for automating MRI picture segmentation. While [10] focus on the contribution of deep learning to improving the classification and accuracy of medical images. Identification of skin lesions, a crucial component of dermatological diagnosis, demonstrates the application of these procedures. Accurate segmentation is essential due to the variety of pigmented skin lesions' benign and malignant appearances. Oliveira et al.'s [11] detailed computational approaches for such segmentation, highlighting the requirement for exact methodologies to successfully distinguish lesion kinds.

Traditional approaches are essential for detecting diseases early, especially now that skin cancer is a prominent cause of death [12], but more recent research has highlighted the difficulties given by the variety of skin lesions [13]. With initial image processing, lesion segmentation, feature extraction methods including GLCM and 2D DWT, and a DNN-based classification as the final step, an integrated approach has obtained an accuracy of 84.45%. Furthermore, the use of pre-trained deep learning models has demonstrated remarkable accuracy rates, with models like MobileNetV2 obtaining up to 98.16% [14], particularly for the diagnosis of specialty diseases like monkeypox. Malignant melanoma is becoming more common, which highlights the need for quick and reliable diagnostic techniques. While traditional methods of feature extraction and preprocessing, like the ABCD rule, have achieved remarkable accuracy of up to 97.8% [15], the advent of deep learning, as demonstrated by the use of VGGNet CNN with transfer learning, offers promising sensitivities of 78.66% for early melanoma detection [16]. However, problems still exist in the field, highlighting the ongoing necessity of improving these diagnostic approaches [17].

With the ability to learn action sequences using Reinforcement Learning (RL) and the robustness of deep learning techniques for representation [18], DRL has become a powerful tool, particularly in the field of medical imaging. The use of DRL in medical imaging includes a variety of operations such as landmark detection, lesion detection, and more [19]. Concurrently, the combination of DRL and CRNN improves feature extraction in multimedia streaming, improving retrieval accuracy and surpassing other annotation models and conventional RL [20]. Interpreting remote sensing images (RSIs) is another use of RL. With an accuracy of more

than 85%, a model using RL in conjunction with gradient-weighted class activation mapping has excellent RSI detection and description accuracy in challenging settings [21]. Use of RL agents for extracting in-depth visual and semantic characteristics has shown to produce improved classification and retrieval results in the field of image classification and retrieval, particularly for intricate historical manuscripts [22]. This development is consistent with RL-based techniques used in other imaging disciplines, such as casting flaw detection, which effectively collaborates via convolutional neural networks and gated recurrent units [23]. Additionally, deep RL's optimization of hyperspectral band selection in remote sensing exhibits better performance, as seen most prominently by a rise in the Kappa coefficient [24]. The use of DRL is further demonstrated in the field of medical imaging, where the DRL-HAPre framework automates picture pre-processing to ensure consistent and improved performance from a variety of sources [25].

Due to high dimensionality and a lack of ground truth data, classification problems in hyperspectral pictures have been solved by developing techniques. One such method combines numerous feature sets for improved accuracy, particularly when training data is limited, by integrating the MDP [26] into its fusion procedures [27]. Concurrently, improvements in the field of autonomous driving have been made thanks to a ground-breaking end-to-end car-following framework. This approach ensures autonomous driving by imitating the leader car's actions using a revised Deep Q-Network algorithm [28], with its effectiveness highlighted in various driving scenarios [29]. A DRL based technique is described, utilising the DDPG [30] for decentralised computation offloading in the quest for decreased power consumption and buffering latency. The DDPG methodology displays greater power allocation capabilities and delivers appreciable reductions in computing costs compared to traditional methods like the deep Q-network (DQN) [31]. There are now numerous attention mechanisms that have substantially advanced the study of image classification [32]–[34]. The Gated Fusion Attention Network (GFANet), which meticulously pulls border features from uneven and hazy lesions, is used to improve skin lesion segmentation and has produced outstanding results on important datasets [35].

III. PROPOSED METHODOLOGY

A. Dataset Description

The ISIC organizes skin lesion challenges every year, motivating more scientists to create CAD tools for lesion detection and furthering skin cancer education. The ISIC 2017 dataset consists of 2750 images: 2000 for training, 150 for validation, and 600 for testing. For accurate lesion segmentation, it's crucial for algorithms to achieve high Specificity (Spe) and Sensitivity (Sen) scores. Notably, during the previous ISIC challenge in 2018, the ground truth for their training data wasn't disclosed. Consequently, the evaluation is primarily focused on the ISIC-2017 dataset. This dataset consists of various images of dermoscopic skin lesions collected from

different dermatoscopes and camera instruments globally. As a result, it's essential to undertake color normalization and illumination adjustments using the color constancy technique. For handling these datasets, the shades of gray algorithm was employed.

B. Proposed Methodology

In the field of medical imaging, precision is crucial, especially when it comes to skin lesion identification. In recognition of this fundamental idea, a cutting-edge, multi-tiered segmentation strategy that is firmly rooted in DRL is presented. The model undergoes continuous learning during each iteration, improving its capabilities by incorporating lessons learned from earlier iterations. The basis of the methodology is the DDPG framework, which is known for understanding the MDP. The actor, who plays a key role in the plan of action, makes decisions by navigating a lean state space, which enables DDPG to create policies that redefine excellence in such models. One of the unique aspects of DDPG is its ERM, which collects learning from a wide range of changes. This quality distinguishes it as the industry benchmark for high-dimensional action landscapes, together with its aptitude for figuring out exact gradients of the action-value function. The incorporation of ResNet-50 and the self-attention mechanism. ResNet-50, as represented in Fig 2., gives more model depth without the usual difficulties with vanishing gradients. By ensuring that the model captures even the finest details in skin photos, this depth lays the groundwork for segmentation accuracy that is unrivaled. On the other hand, the self-attention mechanism improves the focus of the model and emphasizes important details in images. The model's detection abilities are enhanced by this laser-focused focus, which is especially useful for difficult-to-detect skin lesion markers. The attention method ensures a higher sensitivity to conspicuous regions by channeling computational power with delicacy, making it a beacon of accuracy in skin lesion segmentation.

C. Self-Attention Aided ResNet-50 for Actor-Critic Framework

The segmentation of skin lesions is improved by combining the power of DRL with forefront deep learning techniques like ResNet-50 and self-attention. The actor-critic architecture, which is at the core of the system, uses advanced ResNet-50 design to create actors and critics that are not just ordinary neural networks but have the capacity to learn deep representations quickly. The actor-critic mechanism is made up of two networks that are intertwined: the actor, which is responsible for the process of understanding the policy, and the critic, which assesses the activities that the actor suggests. This two-way mechanism makes sure that while the actor aims for an ideal performance, the critic offers a reality check and strives for accurate assessments. The intrinsic interaction between the actor and critic is taken advantage of by this iterative optimization, which is indicative of adversarial networks. The method has been significantly improved by incorporating self-attention mechanisms into the actor and critic ResNet-50 architectures. The attention mechanism enables the networks

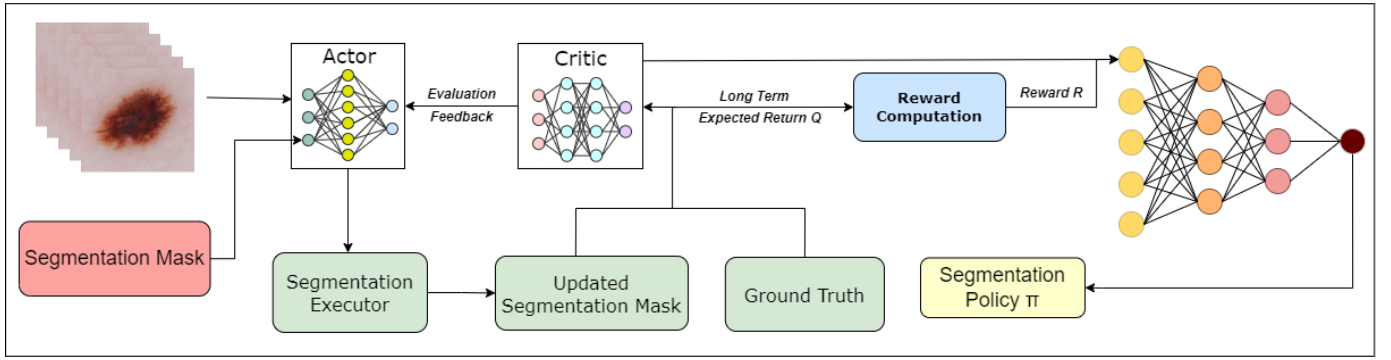


Fig. 2. Proposed RL-based architecture for lesion segmentation task

to prioritize regions that necessitate complex segmentation, allowing them to concentrate on pertinent portions of the input image. This attention-driven methodology makes sure that the network is not only processing images but also attentively analyzing them, adding a level of discernment that is essential for medical image segmentation.

Alongside this, the segmentation executor takes center stage in carrying out the segmentation action and is equipped with a neural network. It creates an action series by using an input image and an initial segmentation mask to establish a new segmentation strategy. The core of this method is a soft tuft brush that expertly creates the texture of each segmentation stroke. This brush is represented as an RL agent. The inclusion of ResNet-50 guarantees that the actor and critic networks gain access to rich, hierarchical features, enhancing the precision and knowledge of their judgments. The networks also employ batch normalization, which speeds up training and serves as a regularizer. The segmentation executor adopts the subpixel technique to fine-tune the brushstroke resolution and produce crisp segmentations. The segmentation executor, enhanced with segmentation techniques, generates diverse segmentation outcomes influenced by the action parameters. Both the actor and critic incorporate the ResNet-50 architecture and self-attention mechanism shown in Fig 3.

D. Enhanced MDP Formulation for Segmentation

The MDP has been redesigned to seamlessly integrate with the sophisticated network architecture, primarily enhanced by ResNet-50 and the attention mechanism. This discussion explains the MDP's refined State, Action, and Reward components within the framework of the ground-breaking segmentation methodology. The state space, which captures how the agent interacts with its surroundings, is the core of the MDP. It is crucial to remember that the state in the novel approach is a sophisticated combination of relevant data and not just a snapshot of the current situation. Specifically, the defined state, represented as $S_t = (S_{mt}, I, t)$, consists of the current segmentation mask S_{mt} , the image I and the step index t . The segmentation mask, S_{mt} , with pixel values oscillating between 0 and 255, distinguishes background pixels (valued at 255) from others. Initially set to 0, this mask

undergoes transformations as the algorithm progresses, with I representing the lesion awaiting segmentation. The step-index t serves as a chronological marker, distinguishing the various stages of segmentation.

In the vast space of potential actions, the agent creates a policy, π , based on the state in the action space. The adjustment of the brushstroke's morphology and direction, which are parameterized in the model for accuracy, is then influenced by this policy. The key concern here is to guarantee that each action the agent takes results in a measurable improvement in the segmentation process. The reward function is the key element, guiding the agent's decisions by mapping states to rewards. The goal remains to augment the cumulative discounted future rewards, R . As the segmentation mask, S_{mt} evolves with each training iteration, its accuracy gets juxtaposed with the ground truth after every step. R_{L2} is introduced as the foundational reward mode by leveraging the L2 mean squared error as a metric. To capture the nuances of R_{L2} across steps, a differential reward function is employed, R_{diff} , defined as Eq. (1).

$$R_{diff} = L2(S_{mt} - 1, G) - L2(S_{mt}, G) \quad (1)$$

When the L2 loss decreases and vice versa, this reward function plays a crucial role in sending signals of affirmation. In order to maintain the consistency of the chosen operation, the reward mechanism also highlights the effectiveness of the action selected over the segmentation span. To appreciate the actions taken by the agent, it becomes pivotal to consider both immediate rewards and long-term benefits. This balance is achieved by the discount factor, γ . The Bellman equation, crucial in establishing the critic's estimation of decisions, is given by Eq. (2) which ensures the critic's estimation augments the segmentation accuracy.

$$Q(S_t, A_t) = R(S_t, A_t) + \gamma \max_{a'} Q(S'_t, A'_t) \quad (2)$$

E. Segmentation Executor and Action Space

The segmentation executor, which is crucial to the strategy, acts as the canvas on which the neural network painstakingly paints brushstrokes to delineate the region of interest (ROI).

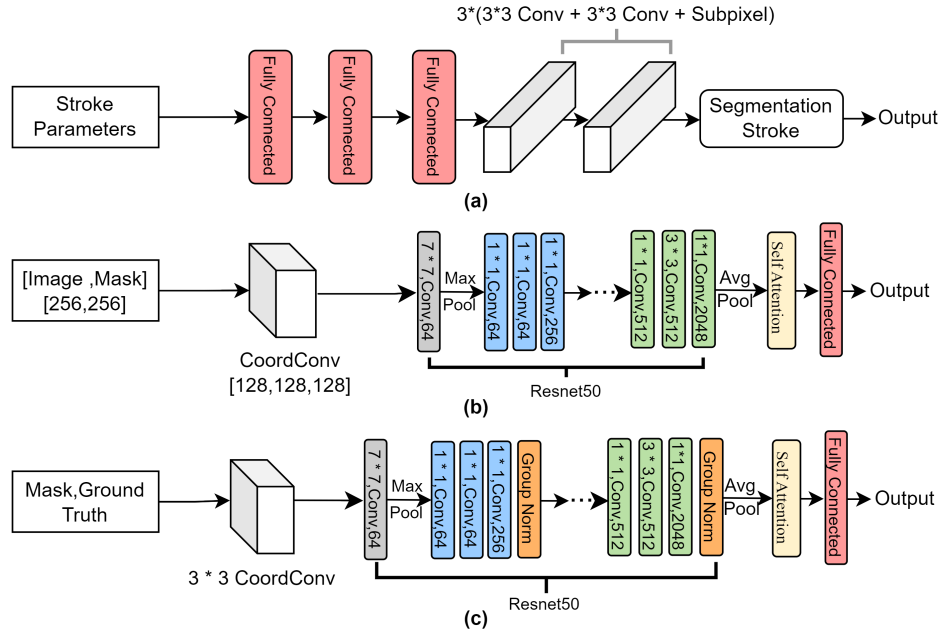


Fig. 3. The network architecture of (a) The segmentation executor, (b) The actor, and (c) The critic

The executor is designed to integrate seamlessly with the DDPG framework. The executor's operating strategy is based on supervised learning algorithms that are fed a wealth of training samples from various graphical rendering systems. A variety of brushstroke shapes, including triangles, circles, and Bézier curves, are created by using various segmentation executors. In contrast to Bezier curves, polynomial B-spline curves stand out for their independence of degree from control points. The flexibility of local control over specific curve sections, guided by control points, is provided by these B-spline curves. QBC was chosen for the segmentation efforts because empirical evidence has shown it to be significantly effective at segmenting lesion images.

For the Quadratic Bezier Curve (QBC), the action parameters are encapsulated in the Eq. (3).

$$A_t = (x_0, y_0, x_1, y_1, x_2, y_2, r_0, r_1), \quad (3)$$

where the control point coordinates for QBC are represented as $(x_0, y_0, x_1, y_1, x_2, y_2)$ or (C_0, C_1, C_2) . The parameters r_0 and r_1 dictate the thickness of QBC endpoints, C_0 and C_2 .

The trajectory traced by a QBC, represented by $B(x)$, is defined in Eq. (4).

$$B(t) = (1 - x)^2 C_0 + 2(1 - x)x C_1 + x^2 C_2, \quad 0 \leq t \leq 1 \quad (4)$$

The frameskip methodology served as the basis for the creative action bundle strategy, which emphasizes segmentation accuracy. A notable enhancement in segmentation accuracy is attained by allowing the actor to formulate an action ensemble, combining multiple actions from the action space, and subsequently permitting the segmentation executor to execute these actions simultaneously.

IV. RESULTS & DISCUSSION

A. Implementation Details

In the experimental setup, an NVIDIA GeForce RTX 2060 GPU with 6GB of VRAM @1350 MHz was utilized to train and evaluate the model. The operating system used was the Arch Linux distribution. A batch size of 32 was used for training and benchmarking, facilitating efficient learning and inference processes. The memory replay experience was set to a value of 500 to ensure robust learning, enhancing the stability of the RL algorithm. For model evaluation, a K-fold validation with K set to 5 was employed, allowing for comprehensive assessment of the model's generalization capabilities. In terms of the RL dynamics, the agent's reward was determined by its interactions with the environment. The agent would receive rewards from the environment based on its actions, and these rewards were instrumental in guiding its learning process. This interaction, akin to a decision-maker

TABLE I
RESULTS COMPARISON OF MODELS ON ISIC 2017 DATASET

Method	DC	Acc	Sen	Spe
U-Net [7]	0.763	0.901	0.672	0.972
SegNet [8]	0.821	0.918	0.801	0.954
Ensemble-S [36]	0.844	0.933	0.805	0.979
Mask R-CNN [37]	0.855	0.935	0.848	0.960
GFANet [35]	0.857	0.939	0.813	0.978
Lina et al. [38]	0.867	0.941	0.892	0.939
SLSNet [39]	0.875	0.944	0.841	0.953
Usmani et al. [5]	0.957	0.954	0.950	0.985
Proposed Model	0.944	0.961	0.941	0.989

TABLE II
EVALUATION RESULTS OF MODELS ON VARIOUS SKIN LESION CATEGORIES

Method	Naevus				Melanoma				Seborrheic Keratosis			
	DC	Acc	Sen	Spe	DC	Acc	Sen	Spe	DC	Acc	Sen	Spe
Ensemble-S [36]	0.879	0.959	0.847	0.979	0.784	0.884	0.733	0.973	0.768	0.899	0.985	0.718
Mask R-CNN [37]	0.888	0.953	0.872	0.963	0.802	0.893	0.786	0.956	0.804	0.908	0.824	0.948
Lina et al. [38]	0.871	0.931	0.872	0.942	0.862	0.926	0.855	0.935	0.843	0.912	0.864	0.941
SLSNet [39]	0.865	0.936	0.862	0.942	0.921	0.925	0.859	0.936	0.861	0.938	0.841	0.942
Usmani et al. [5]	0.930	0.963	0.968	0.986	0.957	0.954	0.939	0.986	0.950	0.942	0.934	0.986
Proposed Model	0.913	0.971	0.975	0.989	0.961	0.963	0.950	0.978	0.935	0.951	0.896	0.965

navigating through a dynamic environment, was a fundamental aspect of the experimental framework, contributing to the learning and adaptation of the RL model.

B. Model Comparison

The proposed models were evaluated against other state-of-the-art models on evaluation metrics including Sen, Spe, Accuracy (Acc), and Dice Coefficient (DC). The DC represents the spatial agreement between the delineated image and the assigned binary mask, as formulated in Eq. (5).

$$DC = \frac{2TP}{2TP + FP + FN} \quad (5)$$

This coefficient is typically employed to gauge shifts in the image axis's transformation center. The terms false positive (FP) and true positive (TP) denote accurately detected lesion pixels and mistakenly tagged healthy pixels as lesions, respectively. Conversely, false negative (FN) highlights lesion pixels that were incorrectly missed while true negative (TN) refers to all inaccurately labeled healthy pixels.

Table 1 presents a comparative analysis between the proposed method and other state-of-the-art techniques. The model notably attains Acc and DC scores of 96.1% and 94.4%, respectively. A closer examination of the qualitative results reveals the superiority of the model's segmentation mask over its counterparts. The comparison of the RL algorithm's statistical metrics with established methods like U-Net, SegNet, and Mask R-CNN, shows its superior capability in discerning intricate lesion patterns, and adapting dynamically to varied skin textures and lesion morphologies. Another notable feature from the results is the enhanced precision of the algorithm in performing segmentation, which consistently demonstrate better performance than those produced by competitor models. This is indicative of the RL algorithm's ability to understand intricate image features and make informed decisions, ensuring accurate lesion segmentation.

A more granular comparison of the RL algorithm is performed across the three lesion categories: naevus, melanoma, and seborrheic keratosis, as presented in Table 2. While the model consistently outperforms in terms of Acc, Sen, and Spe metrics, it exhibits a marginal dip in the DC score specifically for naevus and seborrheic keratosis lesion types. Such deviations can be attributed to intricate morphological variations within these lesion categories. The efficacy of the

proposed RL algorithm is evident in its ability to address the challenges associated with skin lesion identification. One of the defining features of the RL algorithm is its capacity for experiential learning. It adeptly navigates the trade-off between exploration, which delves into discovering the most rewarding strategies to achieve objectives, and exploitation, which evaluates and refines existing optimal solutions based on new insights.

V. CONCLUSION

The addition of a self-attention module suggests a novel multi-phase method for segmenting skin lesions that utilizes a DRL algorithm to increase efficiency. With DRL, an agent is trained to recognise a specific area of interest throughout the segmentation process, which is presented as MDP. This agent sequentially undertakes actions aimed at outlining the target region, with each action characterized by a set of consistent variables. The accuracy of segmentation is further enhanced by refining the replay memory and considering the action set as a tunable parameter. Experimental results suggest that the augmented RL approach delivers commendable outcomes. This strategy holds promise for diversifying into other medical image segmentation domains and varied diagnostic imaging modalities. Notably, the method exhibits prowess in identifying minuscule or irregularly shaped objects, even those without a consistent shape, within the segmentation task. Statistical analyses highlight the superiority of the technique, surpassing leading-edge methods by gains of 94.4% and 96.1% in Dice index and Accuracy, respectively. Consequently, the proposed RL model, enhanced with self-attention, adeptly learns the nuances of segmenting intricate skin lesion visuals.

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